Forest Ecology and Management 549 (2023) 121476

Contents lists available at ScienceDirect

Forest Ecology and Management

journal homepage: www.elsevier.com/locate/foreco

Growth response and survival of American beech, yellow birch, and sugar maple regeneration to partial harvest

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ARTICLE INFO

Keywords:
Hardwoods regeneration
Vegetative reproduction
Partial cut
Soil scarification
Beech control
Deer browsing

ABSTRACT

In northern hardwood stands, silvicultural treatments aimed at regenerating sugar maple and yellow birch, such as site preparation and mechanical control of American beech saplings, often results in the formation of dense thickets of beech regeneration. The objective of this study was to compare the growth and survival of beech regeneration originating from stump sprouts, root suckers and seedlings, with yellow birch and sugar maple regeneration in partially harvested stands under cervid browsing pressure. Over an eight-year period, 997 individuals of the three species were monitored in three northern hardwood stands dominated by sugar maple, yellow birch, and American beech in Quebec, Canada. Each stand included replicates of four partial harvesting treatments and a control, providing a gradient of light and competition conditions. Beech stump sprouts and root suckers had higher growth and survival probabilities than seedlings of all three species under all light and competition conditions. Results suggest that treatments triggering beech sprouts should be applied with caution and under conditions that ensure sufficient survival and development of desired species. This study also provides evidence that selective browsing by cervids could be an important overriding factor exacerbating the effect of beech competition on yellow birch regeneration.

1. Introduction

The establishment of dense American beech (beech, Fagus grandifolia Ehrh.) thickets in the understory of northern hardwood forests of eastern North America is an increasing concern for silviculture and forest management (Bose et al., 2017; Duchesne et al., 2005; Nyland et al., 2006). Beech often dominates the regeneration layer after partial harvest (Rogers et al., 2020) at the expense of other key species such as yellow birch (Betula alleghaniensis Britt.) and sugar maple (Acer saccharum Marsh.). This contrasts with the objectives of current silvicultural regimes in northern hardwood forests, which prioritize the regeneration of both yellow birch and sugar maple to maintain the original stand composition and for their higher economic value (Bernard et al., 2018; Havreljuk et al., 2015).

The abundance of beech in the understory can be explained by its high shade tolerance (Collin et al., 2017), and its ability to reproduce through root suckering and stump sprouting following disturbance, as harvest injuries stimulate adventitious bud initiation (Jones and Raynal, 1988). This can be exacerbated by selective browsing by cervids that target yellow birch and sugar maple over beech (Bose et al., 2018; Matonis et al., 2011). Soil base cations depletion caused by atmospheric acid deposition in the last few decades has also contributed the decline of sugar maple, thereby promoting the expansion of beech in eastern Canada, a species with higher tolerance to acidic soils (Duchesne and Ouimet, 2009).

An important issue associated with beech expansion is that the species is affected by the beech bark disease (BBD), which is caused by the combined action of an insect (Cryptococcus fagisuga or Xylococculus betulae) and a fungal pathogen (Neonectria faginata or N. ditissima) (Cale et al., 2017). Infestation of beech by the BBD challenges the sustainability of these forests as it leads to the degradation of stem quality (Burns and Houston, 1987) and causes significant mortality (Gavin and Peart, 1993; Mize and Lea, 1979), especially in large trees (Houston, 1994). Stand productivity is negatively impacted by the disease (Forrester et al., 2003), thereby reducing the carbon storage capacity of these forests. BBD also reduces the beech mast production, which in turn can affect wildlife as it is an important food resource for some species (Rosemier and Storer, 2010). In turn, the canopy openings created by

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https://doi.org/10.1016/j.foreco.2023.121476
Received 20 June 2023; Received in revised form 8 September 2023; Accepted 2 October 2023
Available online 11 October 2023

overstory beech mortality increase light penetration in the understory and favour the development of beech regeneration (Giencke et al., 2014; Roy and Nolet, 2018). Interventions may then be required to ensure the establishment and development of other species to maintain both the ecological and economic values of northern hardwood forests.

Strategies acting at different levels in the forest structure have been proposed to address the increasing beech density. At the overstory level, larger canopy openings could favour sugar maple as it can perform better than beech under high light conditions (Boulet and Huot, 2013; Dracup and Maclean, 2018; Poulson and Platt, 1996). St-Jean et al. (2021) demonstrated that this strategy can be effective, but only if beech abundance in the overstory is low. When combined with soil scarification, large openings can also promote the establishment of yellow birch, a less shade-tolerant species whose light seeds require mineral soils to germinate (Godman and Krefting, 1960). Yellow birch show higher height growth than beech and sugar maple early after establishment (Bicknell, 1982; Nyland et al., 2004) and react more than these two species increased light conditions (Beaudet and Messier, 1998). However, even when gaps are created by group selection cutting, beech can be the most dominant species in the regeneration (Rogers et al., 2020). At the understory level, silviculture and research efforts have focused on controlling beech regeneration with herbicides, mainly glyphosate, with some success (Kochenderfer et al., 2013, 2004; Nelson and Wagner, 2011). However, forest policies regarding the use of herbicides in forests vary across jurisdiction. For instance, the province of Quebec has banned the use of herbicides in public forests in 2001 (Gouvernement du Québec, 1994), thus highlighting the need for alternative methods. Mechanical control without herbicide application appears to be a valuable alternative (Nyland, 2004; Nyland et al., 2006; Nyland and Kiernan, 2017), but is poorly documented and has shown mixed outcomes. Results from Bédard et al., (2022) showed that the treatment is ineffective to promote the establishment of yellow birch over beech under heavy browsing pressure 10 years after treatment, but the specific response of regeneration to such treatment in terms of growth and survival has not yet been documented.

Previous studies have addressed the development of northern hardwoods regeneration in its very early stages, but little is known about its longer-term response following partial harvest, or of the effects of subsequent treatments such as mechanical control of the understory. Currently, most studies have been conducted in undisturbed stands and have focused on either species comparisons or comparisons of beech root suckers to seedlings with no mention of stump sprouts (Beaudet and Messier, 2008, 1998; Nyland, 2008). Other studies have focused specifically on root suckers (Jones et al., 1989; Jones and Raynal, 1986). However, beech seedlings can establish after a cut and recent evidence challenges the idea that beech thickets originate from root suckers alone (Roy and Nolet, 2018). It is crucial to understand this dynamic, as beech root suckers have been shown to have increased growth and survival compared to seedlings, at least on a short period of time and under closed-canopy conditions (Beaudet and Messier, 2008). Stump sprouts originating from small cut stumps of saplings and poles are also expected to be an important source of beech regeneration after the mechanical control of the beech understory after harvest. Therefore, we lack reliable information to adequately predict the post-harvest developmental trajectory of northern hardwood regeneration following partial cutting and canopy opening.

The aim of this study was to provide a better understanding of the development of beech regeneration compared to yellow birch and sugar maple in harvested stands where partial soil scarification and mechanical control of competing understory species were applied and where cervid browsing pressure was also present. Specifically, we wanted to determine if there was a difference in survival and in both height and diameter growth of beech regeneration (including stump sprouts, root suckers and seedlings) compared to yellow birch and sugar maple seedlings in a gradient of light conditions. We hypothesize that yellow birch has the fastest growth rate early after establishment, especially in high light conditions, and that beech and maple have lower and similar growth under all light conditions. We also hypothesized that beech vegetative reproduction has a higher growth rate than beech seedlings. Finally, we expect the survival rates of beech stump sprouts and root suckers to be higher than those of seedlings from either beech, yellow birch, or sugar maple.

2. Material and methods

2.1. Study area

The study area is located in the Duchesnay Forest (46°57′N, 71°40′W), approximately 35 km northwest of Quebec City, Canada. The Duchesnay Forest is part of the meridional subregion of the balsam fir - yellow birch bioclimatic domain of the Quebec's ecological classification. This subregion includes the northernmost occurrence of key northern hardwood species, such as American beech and sugar maple (Saucier et al., 2009). The climate is subpolar, subhumid, continental, with average annual precipitation ranging between 1200 mm and 1600 mm, and an average growing season of 150 to 170 days (Robitaille and Saucier, 1998). Soils are mainly podzols with moder-type humus derived from a deep glacial till on a gneiss bedrock (Robitaille and Saucier, 1998). Analyses of soil properties revealed deficit in calcium and overall poor fertility throughout the experimental site (Appendix A). The altitude of the experimental site ranges from 230 to 300 m with good to moderate drainage. Depending on the altitude and slope exposition, the forest is mainly composed of yellow birch - balsam fir stands or of sugar maple - yellow birch stands. The composition of the stands selected for the study was dominated by yellow birch, sugar maple and American beech with average merchantable basal areas representing 38 %, 34 % and 23 % of the total basal area of the stands, respectively, before harvest. The Duchesnay Forest is a 8870 ha forest with a history of diameter-limit cutting, just like a large part of the hardwood forests in Quebec (Bédard and Majcen, 2003), and has been affected by the BBD since the early 2010 s. The territory is managed for multiple uses, including an important share of recreational activities and some logging. The white-tailed deer (*Odocoileus virginianus* Zimmerman) density is approximated at 4.02 deer per square kilometer in the larger hunting region where our study site is located (Lebel and De Bellefeuille, 2021), but this density varies significantly within the region. Additionally, hunting is not allowed within the Duchesnay forest, so the density is expected to be higher.

2.2. Experimental design

This study is part of a larger field trial established in 2009 by the *Ministère des Ressources naturelles et des Forêts du Québec* to study multicohort silvicultural scenarios in northern hardwood stands. The experiment described in Bédard et al. (2014) tested four silvicultural treatments a control in four randomized blocks, providing a wide gradient of light and competition conditions to be tested in this study. One block from the original design was not retained for this study as it contains a significantly lower proportion of beech, which was the species of interest in this study. Treatments consisted of a control, a hybrid single-tree and group selection cut with a residual basal area of 18 $m^2 \cdot ha^{-1}$ (SC18), two continuous-cover irregular shelterwood cuts with residual basal area of 16 $m^2 \cdot ha^{-1}$ (CCIS16) and 14 $m^2 \cdot ha^{-1}$ (CCIS14), respectively, and an extended irregular shelterwood (EIS) cut with a residual basal area of 14 $m^2 \cdot ha^{-1}$ (EIS14) (Fig. 1). Blocks were delimited to be homogeneous in terms of forest composition and ecological type as defined by Saucier et al. (2009). In the SC18, CCIS16, and CCIS14 treatments, gaps ranging from 150 to 1000 m^2 were created as part of the cutting treatments. Advance beech and non-commercial regeneration was removed in all gaps using brush saws. Site preparation was also performed in the gaps to remove harvest debris and scarify the soil using

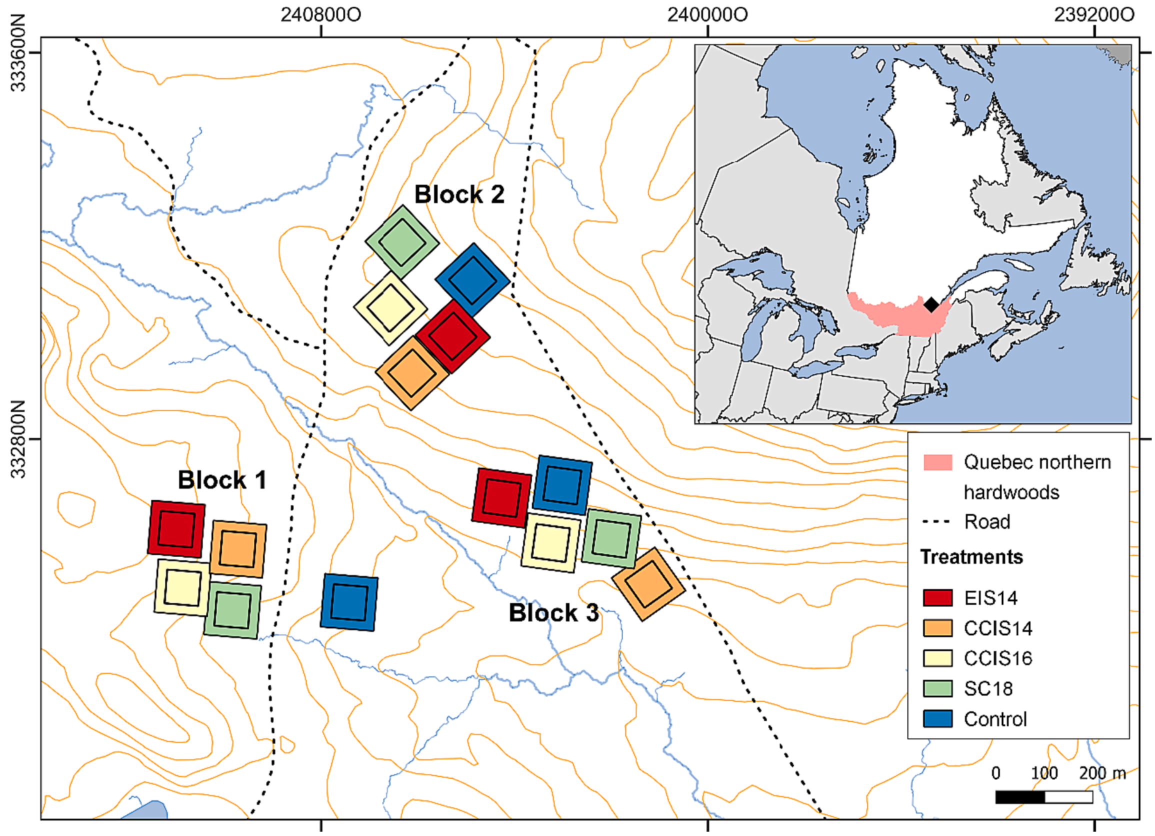

Fig. 1. Study site layout of the treatments and blocks and general study area. Treatments were a control, a hybrid single-tree and group selection cut with a 18 $m^2 \cdot ha^{-1}$ residual basal area (SC18), a continuous-cover irregular shelterwood cut with a residual basal area of 16 $m^2 \cdot ha^{-1}$ (CCIS16), a continuous-cover irregular shelterwood cut with a residual basal area of 14 $m^2 \cdot ha^{-1}$ (CCIS14), and an extended irregular shelterwood cut with a residual basal area of 14 $m^2 \cdot ha^{-1}$ (EIS14).

a fixed-tooth blade equipped on a skidder. In the EIS14 treatment, advance regeneration of beech and non-commercial species was removed throughout the entire experimental units and site preparation was performed between the residual trees. The harvest and post-harvest treatments were conducted during the months of October and November of 2009.

2.3. Data collection

The experimental units were visited three years after the application of the cutting treatments to allow sufficient time for regeneration establishment. The sampling area was a 70 × 70 m central plot within each 110 × 110 experimental unit. Within each plot, 36 circular microplots of 4 m^2 (i.e. 1.13 m radius) were distributed in the plots in a 10 × 10 m systematic grid to conduct regeneration surveys. The initial regeneration survey was conducted 3 years after the treatment and aimed to record seedlings of commercial and main non-commercial species by height class. When sugar maple, yellow birch or beech regeneration was present in a microplot, one representative individual of each species (and origin type for beech) was selected and marked with a unique identifier. Only stems ranging between 6 and 60 cm in height were selected to avoid individuals established before the cut or cotyledons, for which the mortality rate is very high. 152 root suckers, 156 stump sprouts, and 183 seedlings were selected for beech, while 220 and 286 seedlings were selected for yellow birch and sugar maple, respectively, with a total of 997 individuals for the three species. It was generally easy to distinguish between beech root suckers and seedlings as root suckers often originated from exposed roots. Root suckers also had a different form at the collar than seedlings, which are characterized by a tap root. When the origin of the beech was uncertain, the base of the collar was carefully unearthed to determine if the root system was attached to the root system of a parent tree or not. Seedlings were sampled in all treatments (Table 1) to cover a range of environmental conditions, including canopy opening and competition. For each stem, data collected included origin type (seedling, root sprout or stump sprout), living status, presence, or absence of browsing damage, height, diameter at the root collar and the estimated age obtained by counting bud scars. For stump sprouts, the height of the origin of the sprouts from the ground and the diameter of the original stump were also measured. Measurements were repeated every 2 years from 2012 to 2020 for a total of 5 measurements over an 8-year period.

A browsing frequency variable was calculated by cumulating the occurrences of signs of recent browsing at each assessment of the observation period. For example, a seedling that showed signs of recent browsing at measurement times 1, 3, and 4 would have a browsing

Table 1
Total number of tagged seedlings of each species and origin type in the three experimental units for each treatment 3 years after the cut. Treatments were a control, a hybrid single-tree and group selection cut with a 18 $m^2 \cdot ha^{-1}$ residual basal area (SC18), a continuous-cover irregular shelterwood cut with a residual basal area of 16 $m^2 \cdot ha^{-1}$ (CCIS16), a continuous-cover irregular shelterwood cut with a residual basal area of 14 $m^2 \cdot ha^{-1}$ (CCIS14) and an extended irregular shelterwood cut with a residual basal area of 14 $m^2 \cdot ha^{-1}$ (EIS14).

Treatment	Beech (stump sprouts)	Beech (root suckers)	Beech (seedlings)	Yellow birch	Sugar maple	**Total**
Control	12	17	47	8	64	**148**
SC18	30	31	40	44	54	**199**
CCIS16	24	25	28	42	42	**161**
CCIS14	43	39	34	62	62	**240**
EIS14	43	44	33	64	65	**249**
Total	**152**	**156**	**182**	**220**	**287**	**997**

frequency of 3.

In addition to these measurements, a survey of the competing vegetation was conducted in the microplots at the first measurement time, i.e. 3 years after cutting. Seedlings and saplings of woody species (DBH $<$ 1.1 cm), as well as ferns and herbaceous species cover were estimated visually in 10 % classes, and the mean height of each species was also measured. Based on these measurements, a competition index was calculated for each individual:

$$CI = \frac{1}{H_s}\sum_{i=1}^{n} C_i H_i$$

where *CI* is the competition index, Hs is the height of the targeted stem, H_i is the mean height of the i^{th} species in the microplot, and C_i is the ground cover percentage of the i^{th} species in the microplot. The index is independent of the distance to the competitors because any competitor can potentially affect the regeneration growth within the 1.13 m radius of the microplots (Ruel, 1992).

Finally, relative irradiance, i.e. the percentage of above-canopy radiation transmitted to the understory, was estimated in the center of each microplot at 150 cm above ground using hemispherical photographs. Photographs were taken with a Nikon Coolpix 4500 digital camera and a Nikon FC-E8 8 mm fisheye lens before sunrise during the summer of the first and fifth years after cutting. Photographs were analyzed using the Gap Light Analyzer software (Frazer et al., 1999) to calculate the relative irradiance.

2.4. Statistical analysis

A Cox regression model was fitted using the *coxph()* function of the *survival* package (Therneau et al., 2023) to test the effect of different covariates on the probability of survival over an 8-year period. Mortality was treated as a binomial variable taking a value of 1 if a mortality event occurred and of 0 if the tree was still alive at the end of the experiment. Covariates used to predict the survival probability were the species and origin type, initial 3-year relative irradiance, and the initial 3-year competition index. Interactions between species and competition index and between species and relative irradiance were tested. Browsing frequency was not included in the survival model because individuals surviving longer had a higher probability of being browsed at multiple measurement times, which introduces a strong bias in the measure of frequency. The survival model was fitted using the "exact" calculation method as it allows better predictions for interval-censored data, although this method does not allow the inclusion of random effects (Therneau et al., 2023). However, the random effect of the block and experimental unit was investigated beforehand using the same modelling approach, but using the "Efron" calculation method of the *survival* package, which allows for random effects (Therneau et al., 2023), and showed little influence on the model. All 997 individuals were included in the survival analysis.

Two linear mixed effects models were fitted to test the effect of variables of interest on the height and diameter growth of the seedlings, root suckers, and stump sprouts. Linear mixed effects models allow hypotheses to be tested when observations are grouped in multiple levels of experimental units by including both random and fixed effects (Pinheiro and Bates, 2006). Models were built using the *lme()* function of the *nlme* package (Pinheiro et al., 2022) in version 4.2.2 of the R software environment for statistical computing (R Core Team, 2022). Species and origin type (stump sprout, root sucker, or seedling) were included as grouping factors and initial 3-year relative irradiance, and initial 3-year competition index were included as covariates. Interactions between the species (and origin type for beech) and both the competition index and the relative irradiance were also included in the models, as responses to different light intensities may vary between species. A nested random effect on the intercept was included at the experimental unit and block levels to account for the experimental design and the structure of the data. An autoregressive correlation structure was also included in the model to account for repeated measurements on the same individuals. Only stems that survived the 8-year measurement period were included in this model to avoid censored data. Thus, from the initial sample of 997 individuals, 440 were included in the height growth model, i.e. 85 yellow birch seedlings, 60 sugar maple seedlings, 97 beech root suckers, 107 beech stump sprouts, and 91 beech seedlings. The initial runs of the models included the browsing frequency as a covariate, but its effect was inconsistent and non-significant ($p > 0.05$), so we did not include it in the final models.

Model assumptions were checked by examining the residuals for influential outliers or linearity problems. Collinearity between numerical variables was checked using the variance inflation factor (VIF $<$ 3 for all covariates included in the different models) with the *check_collinearity()* function of the *performance* package to avoid including highly correlated variables in the same model. For the linear mixed-effects models, we also assessed the homogeneity of variance and normality of residuals using the *check_model()* function of the *performance* package (Lüdecke et al., 2021). A log transformation was then applied to the response variable of both growth models, i.e. height and diameter, to meet the model's assumptions. The proportional hazards assumption of the Cox regression model was tested using the *cox.zh()* function of the *survival* package.

Model visualisation to investigate the effect of specific parameters was done by predicting values and standard errors while varying the variable of interest, while keeping other covariates of the model at their mean or at a chose value of interest. Predictions were always made for each group, i.e. species and origin type, to examine the differences between them. Pairwise multiple-comparison tests were performed to examine the differences between groups using pairwise comparisons in the *emmeans()* function of the *emmeans* package (Lenth et al., 2023).

3. Results

3.1. Regeneration density

The effects of the treatment showed strong interactions with species as well as origin type for beech with seedlings of the 6 to 60 cm height class three years after the cut. Proportions of beech regeneration originating from stump sprouts and root suckers were highest in the most intensive treatment in terms of percentage of harvest and removal of understory beech, i.e. EIS14 whereas it was the lowest in the control treatment. The three intermediate treatments, i.e. SC18, CCIS16, and CCIS14 showed similar densities of beech vegetative regeneration (Fig. 2).

3.2. Initial characterization of the seedlings

Individuals selected for the follow-up study were representative of different light and competition conditions and were 3 years old in average at the first measurement, i.e. 3 years after the cut (Table 2). Beech stump sprouts originated from residual stumps of an average diameter of 35 mm (sd = 37). The mean height of the stumps was 48 mm aboveground (sd = 85), and the mean height of the origin point of the sprouts on the stumps was 24 mm aboveground (sd = 42).

3.3. Survival probability

The survival probability model showed significant effect of the species, competition index, relative irradiance as well as the interaction between species and competition index on the height of the seedlings, stump sprouts, and root suckers (Table 3). The interaction between species and relative irradiance was also tested, but was not significant ($p = 0.09$) and was not included in the final model. Beech had the highest survival probability 11 years after the cut, with a 0.62 survival probability for stump sprouts 0.60 for root suckers and 0.58 for seedlings.

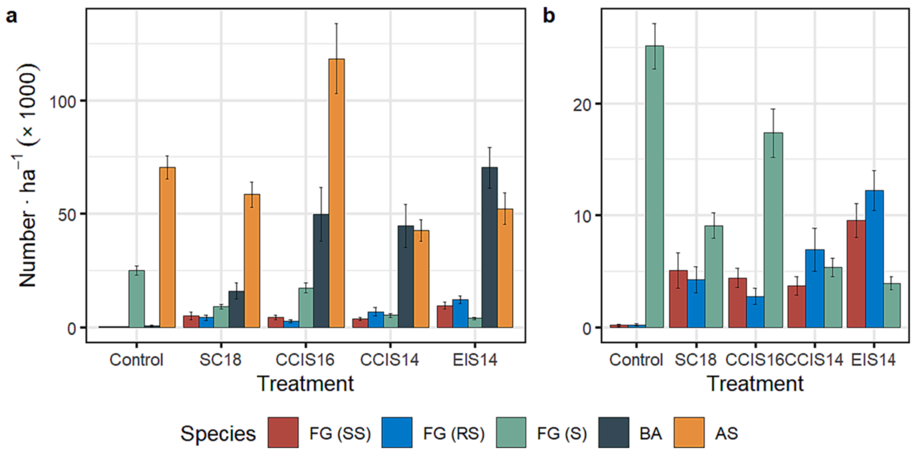

Fig. 2. Mean density of regeneration of the 6 to 60 cm height class of American beech (FG) stump sprouts (SS), root suckers (RS), and seedlings (S), and yellow birch (BA), and sugar maple (AS) seedlings 3 years after the cut (a) in the Duchesnay CPI experiment presented by treatment. Treatments were a control, a hybrid single-tree and group selection cut with a 18 $m^2 \cdot ha^{-1}$ residual basal area (SC18), a continuous-cover irregular shelterwood cut with a residual basal area of 16 $m^2 \cdot ha^{-1}$ (CCIS16), a continuous-cover irregular shelterwood cut with a residual basal area of 14 $m^2 \cdot ha^{-1}$ (CCIS14) and an extended irregular shelterwood cut with a residual basal area of 14 $m^2 \cdot ha^{-1}$ (EIS14). Error bars represent standard errors. Right panel (b) shows a rescale of the density of beech of different origin type only.

Table 2
Initial 3-year characterization of the 997 individuals by species and type of origin 3 years after the cut. Values presented are means ± standard deviation.

Species	Initial age (years)	Initial diameter (mm)	Initial height (mm)	Relative irradiance (%)	Competition index
Beech (stump sprouts)	2.98 ± 0.81	6.95 ± 3.47	437.75 ± 220.06	28.00 ± 10.61	68.89 ± 65.92
Beech (root suckers)	2.76 ± 0.79	5.68 ± 2.71	369.47 ± 185.67	29.94 ± 10.51	83.45 ± 67.65
Beech (seedlings)	3.33 ± 1.74	3.19 ± 1.84	220.70 ± 126.54	23.23 ± 10.40	118.47 ± 122.78
Yellow birch	2.75 ± 0.84	3.40 ± 1.72	260.98 ± 141.72	30.82 ± 9.31	129.02 ± 102.64
Sugar maple	2.32 ± 1.73	2.37 ± 1.52	122.05 ± 91.52	25.32 ± 10.85	240.55 ± 260.79

Table 3
Survival model parameter estimates and their 95 % confidence interval (CI), *z*-value and *p*-value. Yellow birch is included in the intercept. Species are American beech (FG) stump sprouts (SS), root suckers (RS), and seedlings (S), and yellow birch (BA), and sugar maple (AS) seedlings.

Predictors	*Estimates*	*CI*	*z*	*p*
Species				
AS	0.526	0.155—0.897	2.777	**0.006**
FG (RS)	−0.920	−1.465 – -0.376	−3.313	**0.001**
FG (SS)	−1.247	−1.772 – -0.721	−4.650	**<0.001**
FG (S)	−0.726	−1.177 – -0.274	−3.151	**0.002**
Competition	0.001	− 0.000 – 0.003	1.591	0.112
Species*Competition				
AS*Competition	0.001	−0.003—0.001	−0.972	0.331
FG (RS)*Competition	0.003	−0.001—0.007	1.302	0.199
FG (SS)*Competition	0.005	0.001—0.009	2.225	**0.026**
FG (S)*Competition	0.002	0.000—0.004	1.684	0.092
Relative irradiance	−0.027	−0.037 – -0.017	−5.275	**<0.001**

Survival probability was lower for yellow birch at 0.42 and even lower for sugar maple at 0.28 (Fig. 3a). The multiple comparison test showed that beech seedlings from all three origin types had very similar survival probabilities ($p > 0.9$) but differed marginally from yellow birch ($0.09 > p > 0.04$), and that sugar maple differed significantly from all other species ($p < 0.03$). All species and origin types were negatively affected by the competition index, but beech stump sprouts showed the strongest response to increasing competition, followed by beech root suckers. Sugar maple showed almost no change in its survival probability with an increasing competition index (Fig. 3b). Each species showed a higher survival probability with increasing relative irradiance (Fig. 3c). See (Fig. 4).

3.4. Height modelling

The height model showed significant effect of the species, competition index, relative irradiance as well as the interaction between species and competition index and the interaction between species and relative irradiance on the height of the seedlings, stump sprouts and root suckers (Table 4). Beech stump sprouts and root suckers were the tallest for all the studied period and multiple comparison tests showed that they were not significantly different from each other ($p = 0.3$). However, both were significantly different from the other groups ($p < 0.001$). The difference in height between beech seedlings and yellow birch seedlings was also non-significant ($p = 0.99$), but both were significantly different from other groups ($p < 0.001$). Sugar maple seedlings had the lowest height and were significantly different from other groups ($p < 0.001$) (Fig. 3a). Height was negatively affected by the competition index for each species, and beech stump sprouts and root suckers showed the greatest decrease in height with increasing competition. Beech and yellow birch seedlings showed moderate response to increasing competition, while sugar maple seedlings showed practically no response to competition for light (Fig. 3b). For each species and origin type, height increased with the relative irradiance for the interval ranging from 0 to 60 %, except for sugar maple that showed no reaction to increasing irradiance (Fig. 3c).

3.5. Diameter modelling

The diameter model showed a similar pattern to the height model. The diameter model showed significant effect of the species, competition index, relative irradiance as well as the interaction between species and competition index and the interaction term between species and relative irradiance on the height of the seedlings, stump sprouts and root suckers (Table 5). Beech stump sprouts and root suckers had the highest diameter compared to other groups ($p < 0.001$) and were not

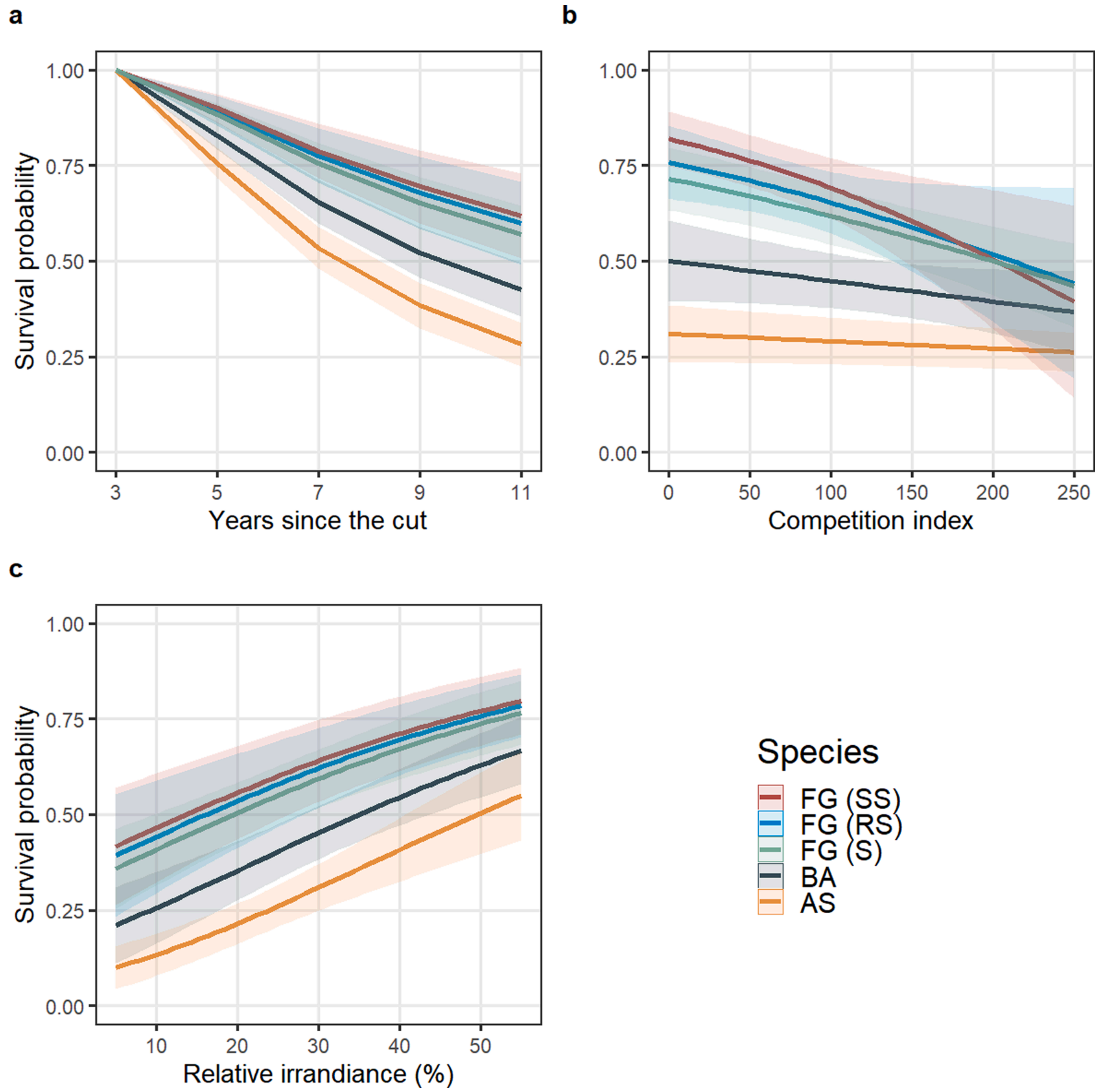

Fig. 3. Predicted survival probabilities and 95 % confidence intervals (shaded area) of American beech (FG) stump sprouts (SS), root suckers (RS), and seedlings (S), and yellow birch (BA), and sugar maple (AS) seedlings on a 8-year period with other parameters kept at their mean values (competition index = 142.79 and relative irradiance = 27.28) (a), effect of the initial 3-year competition index on survival probability 11 years after the cut with relative irradiance kept at its mean value (b) and effect of the initial 3-year relative irradiance on predicted survival probabilities 11 years after the cut with competition index kept at its mean value (c). Predictions were scaled back to the original scale. The survival probabilities plotted here are 1 minus the mortality risk predicitions obtained from the model presented in Table 3.

significantly different from each other ($p = 0.43$). The diameter of beech seedlings and yellow birch seedlings were also not significantly different from each other ($p = 0.99$), and although sugar maple seedlings' diameter was lower, the difference was marginal ($p < 0.08$) (Fig. 5a). Diameter was negatively affected by the competition index for each species, and beech stump sprouts and root suckers showed the greatest negative effect on diameter with increasing competition. Beech seedlings and yellow birch seedlings showed moderate response to increasing competition, while sugar maple seedlings showed practically no response to competition (Fig. 5b). For each species and origin type, diameter increased with the relative irradiance for the interval ranging from 0 to 60 %, but sugar maple showed a much lower response than other species (Fig. 5c).

4. Discussion

4.1. Regeneration density

Partial harvesting followed by mechanical control of the beech understory and soil scarification succeeded in establishing the desired regeneration densities in yellow birch and sugar maple. However, although the total regeneration density of beech was similar between treatments, the density of beech sprouts increased with treatment intensity, while the density of beech seedlings decreased. The increase in beech sprouts density can be explained by two distinct phenomena. First, soil scarification likely damaged and exposed shallow beech roots, which initiated root sprouting (Jones and Raynal, 1988; Ward, 1961). Because beech root suckers usually initiate from shallow roots up to 10 m away from the parent tree (Held, 1983; Jones and Raynal, 1986), it is unlikely that any soil scarification can be done without initiating root sprouting. Outside the canopy gaps, soil was also disturbed in the skid

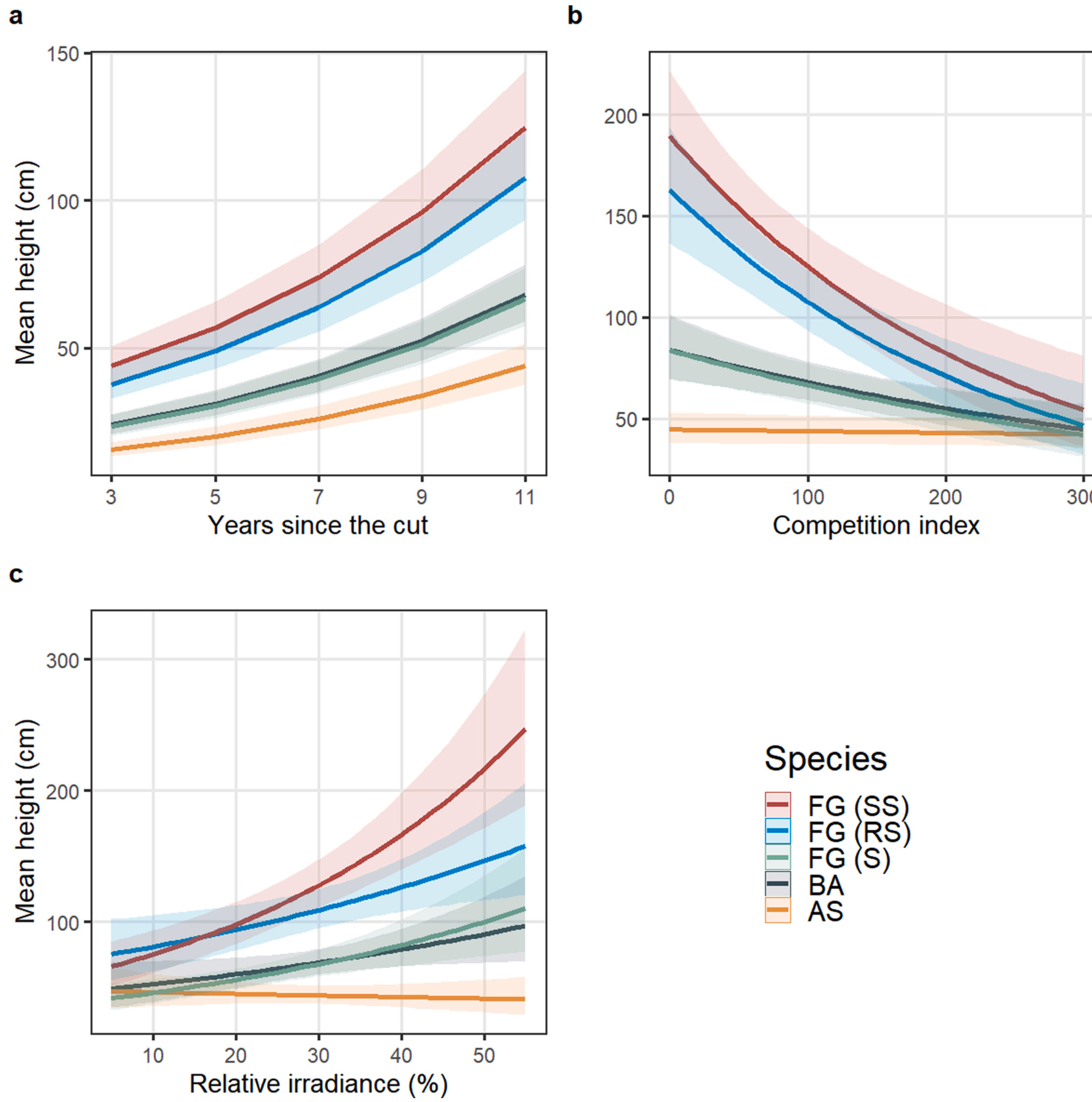

Fig. 4. Predicted height and 95 % confidence intervals (shaded area) of American beech (FG) stump sprouts (SS), root suckers (RS), and seedlings (S), yellow birch (BA), and sugar maple (AS) seedlings on an 8-year period after the cut with other parameters kept at their mean values (competition index = 100.12 and relative irradiance = 29.15) (a), effect of the initial 3-year competition index on predicted height 11 years after the cut with relative irradiance kept at its mean value (b) and effect of the initial 3-year relative irradiance on predicted height 11 years after the cut with competition index kept at its mean value (c). Predictions were scaled back to the original scale.

trails by the circulation of machinery, thus furthering the favourable conditions for sprout initiation. Second, the mechanical control of advanced beech regeneration and harvesting of commercial beech trees initiated stump sprouting. The diameter of the origin stems suggests that stump sprouts mostly originated from small non-commercial stumps created by the mechanical control of the beech understory, which is consistent with what was reported by Nyland et al. (2006). As the treated area increased with increasing treatment intensity, especially in the EIS14 treatment where soil scarification and mechanical control were applied uniformly throughout the stand, the density of vegetative regeneration also increased. These results tend to confirm, at the stand scale, the hypothesis that disturbance by machinery, as well as the mechanical control of the understory, can lead to higher densities of beech root suckers and stump sprouts (Jones and Raynal, 1986; Nyland et al., 2006).

4.2. Survival probability

Beech regeneration of all origin types showed significantly higher survival probabilities than yellow birch and sugar maple seedlings. In fact, sugar maple survival was lower than expected in all tested conditions. Competition, browsing and soil nutrient status are important factors affecting sugar maple seedling survival (Cleavitt et al., 2014, 2011; Hane, 2003), but survival was low even under conditions of low competition. Because the proportion of browsed sugar maple seedlings did not exceed 20 % (Appendix A), we hypothesize that a calcium deficit caused by acidic depositions in the Duchesnay Forest (Houle et al., 1997; Ouimet et al., 2013) limited sugar maple survival without affecting yellow birch and beech survival (Kobe et al., 2002). Despite the absence of evidence in our analyses, yellow birch survival may have been affected by browsing (Elie et al., 2009).

Contrary to our initial hypothesis, we did not observe any significant differences in the survival probabilities of beech from different origins. Results from Beaudet and Messier (2008) showed that beech sprouts had

Table 4
Height model parameter estimates and their 95 % confidence interval (CI), *t*-value and *p*-value. Yellow birch is included in the intercept. Time corresponds to the years since the cut. Species are American beech (FG) stump sprouts (SS), root suckers (RS), and seedlings (S), and yellow birch (BA) and sugar maple (AS) seedlings.

Predictors	*Estimates*	*CI*	*t*	*p*
Intercept	4.892	4.494 – 5.290	24.106	**<0.001**
Time	0.131	0.123 – 0.139	31.012	**<0.001**
Species				
AS	−0.148	−0.668 – 0.373	−0.557	0.578
FG (RS)	0.632	0.135 – 1.130	2.494	**0.013**
FG (SS)	0.443	−0.036 – 0.923	1.814	0.070
FG (S)	−0.167	−0.641 – 0.306	−0.693	0.487
Competition	−0.002	−0.003 – - 0.001	−3.758	<0.001
Species*Competition				
AS*Competition	0.002	0.001 – 0.003	3.250	**0.001**
FG (RS)*Competition	−0.002	−0.004 – - 0.000	−2.170	0.030
FG (SS)*Competition	−0.002	−0.004 – - 0.000	−2.124	**0.034**
FG (S)*Competition	−0.000	−0.002 – 0.001	−0.252	0.801
Relative irradiance	0.014	0.001 – 0.026	2.189	0.029
Species*Relative irradiance				
AS*Relative irradiance	−0.016	−0.033 – 0.000	−1.928	0.054
FG (RS)*Relative irradiance	0.001	−0.015 – 0.017	0.136	0.892
FG (SS)*Relative irradiance	0.013	−0.002 – - 0.028	1.685	0.092
FG (S)*Relative irradiance	0.006	−0.010 – 0.022	0.704	0.481

Table 5
Diameter model parameter estimates and their 95 % confidence interval (CI), *t*-value and *p*-value. Yellow birch is included in the intercept. Species are American beech (FG) stump sprouts (SS), root suckers (RS), and seedlings (S), and yellow birch (BA), and sugar maple (AS) seedlings.

Predictors	*Estimates*	*CI*	*t*	*p*
Intercept	0.433	0.109 – 0.756	2.620	**0.009**
Time	0.122	0.117 – 0.127	48.977	**<0.001**
Species				
AS	0.215	−0.212 – - 0.642	0.987	0.324
FG (RS)	0.803	0.395 – 1.211	3.862	**<0.001**
FG (SS)	0.609	0.216 – 1.002	3.038	**0.002**
FG (S)	0.008	−0.380 – 0.397	0.042	0.967
Competition	−0.002	−0.003 – - 0.001	−3.590	<0.001
Species*Competition				
AS*Competition	0.001	−0.000 – 0.002	2.608	**0.009**
FG (RS)*Competition	−0.002	−0.003 – 0.000	−2.042	0.041
FG (SS)*Competition	−0.001	−0.003 – 0.000	−1.670	0.095
FG (S)*Competition	−0.000	−0.002 – 0.001	−0.294	0.769
Relative irradiance	0.021	0.011 – 0.032	8.751	**<0.001**
Species*Relative irradiance				
AS*Relative irradiance	−0.019	−0.032 – 0.005	−2.643	**0.008**
FG (RS)* Relative irradiance	−0.007	−0.019 – 0.006	−0.994	0.320
FG (SS)* Relative irradiance	−0.003	−0.009 – 0.015	0.481	0.631
FG (S)* Relative irradiance	−0.000	−0.013– 0.013	−0.024	0.981

higher survival rates than seedlings under closed canopy conditions, while Houston (2001) observed higher survival rates for seedlings in partial cuts and clearcuts. This could indicate that sprouts are more vigorous than seedlings when resources are limited, due to their connection to a parent tree, but that this advantage is not as crucial for survival in a situation where resources are not limiting, i.e. after canopy opening. After harvest, parent trees are also likely to be cut, which has also been shown to reduce survival rates for sprouts (Houston, 2001). Overall, no light or competition conditions tested in this study were sufficient to promote the survival of yellow birch or sugar maple over that of beech (Fig. 3).

4.3. Height and diameter growth

The higher growth rate of beech vegetative reproduction compared to beech seedlings confirms our hypothesis and is consistent with a previous study showing that small beech root suckers (<30 cm) have higher initial growth rates under closed canopy conditions (Beaudet and Messier, 2008). Our results confirm this trend over a longer observation period and under a gradient of light and competition conditions. They also confirm that this trend is applicable to stump sprouts. Beech vegetative reproduction was on average taller than that of seedlings of all three species 11 years after the cut, whereas beech seedlings were of a similar height to those of yellow birch, and would most likely not be as problematic for the growth and survival of other species. Yellow birch height growth was lower than we hypothesized, suggesting an important overriding effect of browsing pressure by cervids, as evidenced by Elie et al. (2009). The presence of white-tailed deer was confirmed by camera traps installed in the experimental units during this study. Yellow birch was also clearly preferred by cervids in this study, as we observed that up to 60 % of the yellow birch seedlings had browsing damage, while proportions were of less than 20 % an 10 % for sugar maple and beech, respectively (Appendix A). Other studies have noted significant browsing damage to sugar maple (Bose et al., 2018; Matonis et al., 2011), but this may be relative to the availability of preferred species. Our results are in line with those of Kittredge and Ashton (1995) where birch was selected over sugar maple.

Despite the fact that browsing has been shown to be an important driver of regeneration dynamics in northern hardwood forests (Bédard et al., 2022; Bose et al., 2018; Sage et al., 2003), we could not account for its effect in this study. This was most likely attributable to the way browsing was monitored as a binary variable, i.e. absence or presence of browsing damage at each measurement to calculate a browsing frequency. Instead, quantifying the severity of browsing pressure at the site level should be considered in future studies to account for this effect (Keigley and Frisina, 2011). Exclosure experiments are also important sources of information to isolate the browsing impact on seedlings growth (ex: Kern et al., 2012), although they may not be representative of the actual regeneration success that may be achieved in practice.

The growth rate of sugar maple was also lower than we hypothesized, which is unlikely to be entirely explained by browsing as it was relatively low for this species (Appendix A). Instead, it may have been affected by a calcium limitation offsetting the effect of partial canopy opening for sugar maple while not being detrimental to beech growth (Duchesne et al., 2013; Kobe et al., 2002; Nolet et al., 2008). It is also possible that the high initial density of yellow birch and beech seedlings limited early growth of sugar maple. However, no light or competition conditions tested in this study were sufficient to promote the growth of yellow birch or sugar maple so it can outcompete that of beech stump sprouts or root suckers.

4.4. Implications for management

Our results show that the combination of low mortality and high growth rates of beech vegetative reproduction compared to both yellow birch and sugar maple seedlings can allow beech to dominate the sapling layer 11 years after partial cut. They also show that no light or competition conditions found after partial harvest were sufficient to sugar maple or yellow birch seedlings at the expense of beech clones. These results confirm that vegetative reproduction of beech plays an important role in the community dynamics of northern hardwood forests, and that these dynamics should not be reduced to between-species interactions. The proportion of vegetative reproduction shows considerable variations among stands (Nyland, 2008). Our results confirm that silvicultural factors such as intensive partial harvest treatments including soil scarification and mechanical control of the advanced beech regeneration

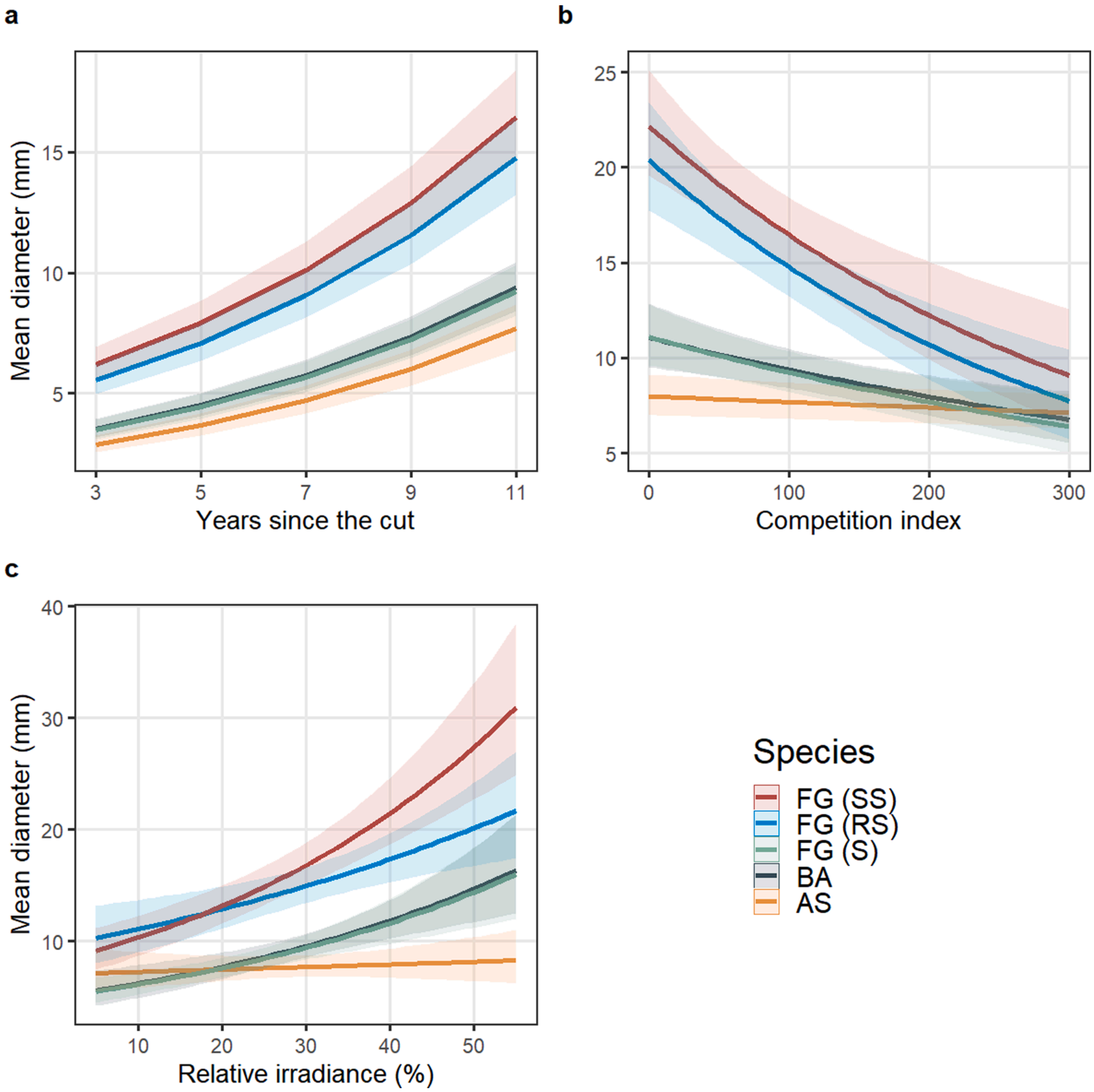

Fig. 5. Predicted diameter and 95 % confidence intervals (shaded area) of American beech (FG) stump sprouts (SS), root suckers (RS), and seedlings (S), yellow birch (BA), and sugar maple (AS) seedlings on an 8-year period after the cut with other parameters kept at their mean values (competition index = 100.12 and relative irradiance = 29.15) (a), effect of the initial 3-year competition index on predicted diameter 11 years after the cut with relative irradiance kept at its mean value (b) and effect of the initial 3-year relative irradiance on predicted diameter 11 years after the cut with competition index kept at its mean value (c). Predictions were scaled back to the original scale.

had the effect of promoting abundant and vigorous beech stump sprouts and root suckers. Finally, site factors have also been shown to affect the abundance of beech vegetative reproduction (Takahashi et al., 2010).

This study demonstrates that increasing understory light penetration by creating canopy gaps or increased level of harvest to favour sugar maple or yellow birch over beech (Dracup and Maclean, 2018; Poulson and Platt, 1996), and the removal of beech advance regeneration (Nolet et al., 2015; Nyland et al., 2006, 2004), is not necessarily sufficient to secure a regeneration success. The regeneration window is typically of only a few years in temperate forests (Beaudet et al., 2007), and promoting good growth and survival conditions for the regeneration during this short period is critical. The strong growth response of beech sprouts combined with cervid browsing pressure can promote beech dominance after partial harvest. This study has shown that the effect of a mechanical control of understory beech can be completely offset if the conditions are not met to ensure the growth and survival of the desired species, supporting similar results of Bose et al. (2018) with chemical control methods. Therefore, treatments to control the beech understory should be carried out with caution and should be applied with careful consideration of the risk of stimulating vegetative reproduction. In some cases, it may also be necessary to implement repeated mechanical control interventions to ensure the proper development of the desired species (Direction de la recherche forestière, 2017).

Sugar maple showed little response in both growth and survival to increased light or reduced competition, indicating that the species does not necessarily benefit from larger openings unless it already dominates the regeneration layer. This result is consistent with the findings of Beaudet et al. (2007) and St-Jean et al. (2021). In specific cases where soil calcium deficit is identified as a limiting factor for sugar maple seedling development, liming could be considered to improve its chances to compete with beech regeneration (Bognounou et al., 2023; Duchesne et al., 2013), but this strategy has shown mixed results (Nolet et al., 2015) and is likely only applicable for important calcium deficit. In any case, a better understanding of the conditions favouring the regeneration success of the desired species is necessary before implementing strategies based on the mechanical control of the beech advance regeneration as applied in this study. Soil base cation status, browsing by cervids and the height and spatial dispersion of the established regeneration are likely to influence the success of such treatments. These effects will need to be documented to facilitate the implementation of bespoke silvicultural prescriptions that will favour regeneration success in northern hardwood forests.

5. Conclusions

Our results showed that even when cohorts of yellow birch and sugar maple establish properly following different intensities of partial harvest and post-harvest treatments in northern hardwood stands, vegetative reproduction of beech can outcompete them. Beech sprouts and suckers outcompeted beech seedlings as well as those of other species in both height and diameter growth, but also showed increased survival probabilities over time. Beech clones were able to outcompete other species even when treatment intensity was high, inducing more sprouting from injured roots and cut stumps. However, we also observed that yellow birch seedlings were severely affected by heavy deer browsing, which certainly affected their regeneration success. When partial harvesting is aimed at establishing a new cohort of desired species, large openings and post-harvest treatments may not always be an appropriate tool and may even exacerbate the problem of beech expansion if not prescribed with caution and consideration of site-specific factors such as the status of the established regeneration in desired species, the risk of heavy browsing by cervids and the soil nutritive status. The effects of beech stump sprouts and root suckers on yellow birch and sugar maple seedlings highlight the need for both fine-scale silvicultural interventions and holistic management strategies to maintain these two species in the northern hardwood forests of North America, and thereby maintain the ecological and economic sustainability of these forests.

CRediT authorship contribution statement

Sébastien Dumont: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing. **Steve Bédard:** Conceptualization, Supervision, Methodology, Writing – review & editing, Funding acquisition. **Alexis Achim:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors are grateful to Patricia Raymond, Stéphane Tremblay, Catherine Larouche, Vincent Roy, and Marilou Beaudet for their expertise in developing the experimental design and data collection protocol. Special thanks to the hardwood silviculture technical team of the Direction de la recherche forestière for their hard work in the field and observations on the experiment, and to Josianne DeBlois, Catherine Chagnon and Guillaume Moreau for their advice on statistical analyses. This research is part of the first author's master's thesis at Université Laval and was supported by a Produits forestiers D&G scholarship and the Ministère des Ressources naturelles et des Forêts. This work is part of project number 142332022 conducted at the Direction de la recherche forestière (ministère des Ressources naturelles et des Forêts, Québec, Canada) and led by Steve Bédard.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foreco.2023.121476.

References

Beaudet, M., Brisson, J., Gravel, D., Messier, C., 2007. Effect of a major canopy disturbance on the coexistence of acer saccharum and fagus grandifolia in the understorey of an old-growth forest. J. Ecol. 95, 458–467. https://doi.org/10.1111/j.1365-2745.2007.01219.x.

Beaudet, M., Messier, C., 1998. Growth and morphological responses of yellow birch, sugar maple, and beech seedlings growing under a natural light gradient. Can. J. For. Res. 28, 1007–1015. https://doi.org/10.1139/X98-077.

Beaudet, M., Messier, C., 2008. Beech regeneration of seed and root sucker origin: a comparison of morphology, growth, survival, and response to defoliation. For. Ecol. Manage. 255, 3659–3666. https://doi.org/10.1016/j.foreco.2008.03.010.

Bédard, S., Guillemette, F., Raymond, P., Tremblay, S., Larouche, C., DeBlois, J., 2014. Rehabilitation of northern hardwood stands using multicohort silvicultural scenarios in Québec. J. For. 112, 276–286. https://doi.org/10.5849/jof.13-035.

Bédard, S., Majcen, Z., 2003. Growth following single-tree selection cutting in Québec northern hardwoods. For. Chron. 79, 898–905. https://doi.org/10.5558/tfc79898-5.

Bédard, S., Raymond, P., DeBlois, J., 2022. Northern hardwood regeneration dynamics 10 years after irregular shelterwood and mechanical control of understory American beech. For. Ecol. Manage. 511, 120142 https://doi.org/10.1016/j.foreco.2022.120142.

Bernard, A., Gélinas, N., Duchateau, E., Durocher, C., Achim, A., 2018. American beech in value-added hardwood products: assessing consumer preferences. BioResources 13. https://doi.org/10.15376/biores.13.3.6893-6910.

Bicknell, S.H., 1982. Development of canopy stratification during early succession in northern hardwoods. For. Ecol. Manage. 4, 41–51. https://doi.org/10.1016/0378-1127(82)90027-5.

Bognounou, F., Paré, D., Laganière, J., 2023. Changes in seedlings' composition and abundance following soil scarification and amendments in a northern hardwood forest. For. Ecol. Manage. 541, 121071 https://doi.org/10.1016/j.foreco.2023.121071.

Bose, A.K., Weiskittel, A., Wagner, R.G., 2017. A three decade assessment of climate-associated changes in forest composition across the north-eastern USA. J. Appl. Ecol. 54, 1592–1604. https://doi.org/10.1111/1365-2664.12917.

Bose, A.K., Wagner, R.G., Roth, B.E., Weiskittel, A.R., 2018. Influence of browsing damage and overstory cover on regeneration of American beech and sugar maple nine years following understory herbicide release in central Maine. New For. 49, 67–85. https://doi.org/10.1007/s11056-017-9605-5.

Boulet, B., Huot, M., 2013. Le guide sylvicole du Québec - Tome I : Les fondements biologiques de la sylviculture. Publications du Québec, Publications du Québec, Québec.

Burns, B.S., Houston, D.R., 1987. Managing beech bark disease: evaluating defects and reducing losses. North. J. Appl. For. 4, 28–33. https://doi.org/10.1093/njaf/4.1.28.

Cale, J.A., Garrison-Johnston, M.T., Teale, S.A., Castello, J.D., 2017. Beech bark disease in North America: over a century of research revisited. For. Ecol. Manage. 394, 86–103. https://doi.org/10.1016/j.foreco.2017.03.031.

Cleavitt, N.L., Fahey, T.J., Battles, J.J., 2011. Regeneration ecology of sugar maple (Acer saccharum): seedling survival in relation to nutrition, site factors, and damage by insects and pathogens. Can. J. For. Res. 41, 235–244. https://doi.org/10.1139/X10-210.

Cleavitt, N.L., Battles, J.J., Fahey, T.J., Blum, J.D., 2014. Determinants of survival over 7 years for a natural cohort of sugar maple seedlings in a northern hardwood forest. Can. J. For. Res. 44, 1112–1121. https://doi.org/10.1139/cjfr-2014-0177.

Collin, A., Messier, C., Kembel, S.W., Bélanger, N., 2017. Low Light Availability Associated with American Beech Is the Main Factor for Reduced Sugar Maple Seedling Survival and Growth Rates in a Hardwood Forest of Southern Quebec. Forests 2017, Vol. 8, Page 413 8, 413. https://doi.org/10.3390/F8110413.

Direction de la recherche forestière, 2017. Expansion du hêtre à grandes feuilles et déclin de l'érable à sucre au Québec : portrait de la situation, défis et pistes de solution (Avis scientifique). Gouvernement du Québec, ministère des Forêts, de la Faune et des Parcs, Direction de la recherche forestière, Québec.

Dracup, E.C., Maclean, D.A., 2018. Partial harvest to reduce occurrence of American beech affected by beech bark disease: 10 year results. Forestry 91, 73–82. https://doi.org/10.1093/forestry/cpx033.

Gouvernement du Québec, 1994. La Stratégie de protection des forêts. Ministère des Forêts, de la Faune et des Parcs.

Duchesne, L., Moore, J.-D., Ouimet, R., 2013. Partitioning the effect of release and liming on growth of sugar maple and american beech saplings. North. J. Appl. For. 30, 28–36. https://doi.org/10.5849/njaf.12-012.

Duchesne, L., Ouimet, R., 2009. Present-Day Expansion of American Beech in Northeastern Hardwood Forests: Does Soil Base Status Matter? 39, 2273–2282. https://doi.org/10.1139/X09-172. https://doi.org/10.1139/X09-172.

Duchesne, L., Ouimet, R., Moore, J.-D., Paquin, R., 2005. Changes in structure and composition of maple–beech stands following sugar maple decline in Québec, Canada. For. Ecol. Manage. 208, 223–236. https://doi.org/10.1016/j.foreco.2004.12.003.

Elie, J.-G., Ruel, J.-C., Lussier, J.-M., 2009. Effect of browsing, seedbed, and competition on the development of yellow birch seedlings in high-graded stands. North. J. Appl. For. 26, 99–105. https://doi.org/10.1093/njaf/26.3.99.

Forrester, J.A., McGee, G.G., Mitchell, M.J., 2003. Effects of beech bark disease on aboveground biomass and species composition in a mature northern hardwood forest, 1985 to 2000. J. Torrey Bot. Soc. 130, 70–78. https://doi.org/10.2307/3557531.

Frazer, G.W., Canham, C.D., Lertzman, K.P., 1999. Gap light analyser, version 2.0: Imaging software to extract canopy structure and gap light indices from true-colour fisheye photographs.

Gavin, D.G., Peart, D.R., 1993. Effects of beech bark disease on the growth of American beech (Fagus grandifolia). Can. J. For. Res. 23, 1566–1575. https://doi.org/10.1139/x93-197.

Giencke, L.M., Dovčiak, M., Mountrakis, G., Cale, J.A., Mitchell, M.J., 2014. Beech bark disease: spatial patterns of thicket formation and disease spread in an aftermath forest in the northeastern United States. Can. J. For. Res. 44, 1042–1050. https://doi.org/10.1139/cjfr-2014-0038.

Godman, R.M., Krefting, L.W., 1960. Factors important to yellow birch establishment in upper michigan. Ecology 41, 18–28. https://doi.org/10.2307/1931935.

Hane, E.N., 2003. Indirect effects of beech bark disease on sugar maple seedling survival. Can. J. For. Res. 33, 807. https://doi.org/10.1139/x03-008.

Havreljuk, F., Bédard, S., Guillemette, F., DeBlois, J., 2015. Predicting log grade volumes in northern hardwood stands of southern Quebec, in: Proceedings—ISCHP 2015: 5th International Scientific Conference on Hardwood Processing. Québec, pp. 107–114.

Held, M.E., 1983. Pattern of beech regeneration in the East-Central United States. Bull. Torrey Bot. Club 110, 55–62. https://doi.org/10.2307/2996517.

Houle, D., Paquin, R., Camiré, C., Ouimet, R., Duchesne, L., 1997. Response of the Lake Clair Watershed (Duchesnay, Quebec) to changes in precipitation chemistry (1988–1994). Can. J. For. Res. 27, 1813–1821. https://doi.org/10.1139/x97-143.

Houston, D.R., 1994. Major new tree disease epidemics: beech bark disease. Annu. Rev. Phytopathol. 32, 75–87. https://doi.org/10.1146/annurev.py.32.090194.000451.

Houston, D.R., 2001. Effect of harvesting regime on beech root sprouts and seedlings in a north-central Maine forest long affected by beech bark disease. Res. Pap. NE-717. Newtown Square, PA: U.S. Department of Agriculture, Forest Service, Northeastern Research Station. 20 p. 717. https://doi.org/10.2737/NE-RP-717.

Jones, R.H., Raynal, D.J., 1988. Root sprouting in American beech (Fagus grandifolia): effects of root injury, root exposure, and season. For. Ecol. Manage. 25, 79–90. https://doi.org/10.1016/0378-1127(88)90121-1.

Jones, R.H., Nyland, R.D., Raynal, D.J., 1989. Response of American beech regeneration to selection cutting of Northern Hardwoods in New York. North. J. Appl. For. 6, 34–36. https://doi.org/10.1093/njaf/6.1.34.

Jones, R.H., Raynal, D.J., 1986. Spatial distribution and development of root sprouts in Fagus grandifolia (Fagaceae). Am. J. Bot. 73, 1723. https://doi.org/10.2307/2444238.

Keigley, R.B., Frisina, M.R., 2011. Process to Monitor and Manage Ungulate Browsing Pressure 16.

Kern, C.C., Reich, P.B., Montgomery, R.A., Strong, T.F., 2012. Do deer and shrubs override canopy gap size effects on growth and survival of yellow birch, northern red oak, eastern white pine, and eastern hemlock seedlings? For. Ecol. Manage. 267, 134–143. https://doi.org/10.1016/j.foreco.2011.12.002.

Kittredge, D.B., Ashton, P.M.S., 1995. Impact of deer browsing on regeneration in mixed stands in Southern New England. North. J. Appl. For. 12, 115–120. https://doi.org/10.1093/njaf/12.3.115.

Kobe, R.K., Likens, G.E., Eagar, C., 2002. Tree seedling growth and mortality responses to manipulations of calcium and aluminum in a northern hardwood forest. Can. J. For. Res. 32, 954–966. https://doi.org/10.1139/x02-018.

Kochenderfer, J.D., Kochenderfer, J.N., Warner, D.A., Miller, G.W., 2004. Preharvest manual herbicide treatments for controlling American Beech in Central West Virginia. North. J. Appl. For. 21, 40–49. https://doi.org/10.1093/njaf/21.1.40.

Kochenderfer, J.D., Kochenderfer, J.N., Miller, G.W., 2013. Extending the time interval for applying herbicide in cut-stump treatments on American Beech. North. J. Appl. For. 30, 118–124. https://doi.org/10.5849/njaf.12-033.

Lebel, F., De Bellefeuille, S., 2021. Plan de gestion du cerf de Virginie au Québec 2020-2027. Direction de l'expertise sur la faune terrestre, l'herpétofaune et l'avifaune, Direction générale de la gestion de la faune et des habitats, ministère des Forêts, de la Faune et des Parcs, Québec.

Lenth, R.V., Buerkner, P., Giné-Vázquez, I., Herve, M., Jung, M., Love, J., Miguez, F., Riebl, H., Singmann, H., 2023. emmeans: Estimated Marginal Means, aka Least-Squares Means.

Lüdecke, D., Ben-Shachar, M.S., Patil, I., Waggoner, P., Makowski, D., 2021. performance: an R package for assessment, comparison and testing of statistical models. Journal of Open Source Software 6, 3139. https://doi.org/10.21105/joss.03139.

Matonis, M.S., Walters, M.B., Millington, J.D.A., 2011. Gap-, stand-, and landscape-scale factors contribute to poor sugar maple regeneration after timber harvest. For. Ecol. Manage. 262, 286–298. https://doi.org/10.1016/j.foreco.2011.03.034.

Mize, C.W., Lea, R.V., 1979. The effect of the beech bark disease on the growth and survival of beech in northern hardwoods. For. Pathol. 9, 242–248. https://doi.org/10.1111/j.1439-0329.1979.tb00684.x.

Nelson, A.S., Wagner, R.G., 2011. Improving the composition of beech-dominated northern hardwood understories in Northern Maine. North. J. Appl. For. 28, 186–193. https://doi.org/10.1093/njaf/28.4.186.

Nolet, P., Bouffard, D., Doyon, F., Delagrange, S., 2008. Relationship between canopy disturbance history and current sapling density of Fagus grandifolia and Acer saccharum in a northern hardwood landscape. Can. J. For. Res. 38, 216–225. https://doi.org/10.1139/X07-160.

Nolet, P., Delagrange, S., Bannon, K., Messier, C., Kneeshaw, D., 2015. Liming has a limited effect on sugar maple – American beech dynamics compared with beech sapling elimination and canopy opening. Can. J. For. Res. 45, 1376–1386. https://doi.org/10.1139/cjfr-2015-0010.

Nyland, R.D., 2004. Simple girdle kills small American beech (Fagus grandifolia Ehrh.). North. J. Appl. For. 21, 220–221. https://doi.org/10.1093/njaf/21.4.220.

Nyland, R.D., 2008. Origin of small understory beech in New York northern hardwood stands. North. J. Appl. For. 25, 161–163. https://doi.org/10.1093/njaf/25.3.161.

Nyland, R.D., Bashant, A.L., Bohn, K.K., Verostek, J.M., 2006. Interference to hardwood regeneration in northeastern North America: controlling effects of American Beech, Striped Maple, and Hobblebush. North. J. Appl. For. 23, 122–132. https://doi.org/10.1093/njaf/23.2.122.

Nyland, R.D., Kiernan, D.H., 2017. Beech shearing/high-stumping to control understory interference. For. Chron. 93, 199–203. https://doi.org/10.5558/tfc2017-028.

Nyland, R.D., Ray, D.G., Yanai, R.D., 2004. Height development of upper-canopy trees within even-aged adirondack northern hardwood stands. North. J. Appl. For. 21, 117–122. https://doi.org/10.1093/njaf/21.3.117.

Ouimet, R., Moore, J.D., Duchesne, L., 2013. Soil thresholds update for diagnosing foliar calcium. Potassium, or Phosphorus Deficiency of Sugar Maple 44, 2408–2427. https://doi.org/10.1080/00103624.2013.803563.

Pinheiro, J., Bates, D., 2006. Mixed-Effects Models in S and S-PLUS. Springer Science & Business Media.

Pinheiro, J., Bates, D., DebRoy, S., Sarkar, D., R Core Team, 2022. nlme: Linear and Nonlinear Mixed Effects Models.

Poulson, T.L., Platt, W.J., 1996. Replacement patterns of beech and sugar maple in warren woods, Michigan. Ecology 77, 1234–1253. https://doi.org/10.2307/2265592.

R Core Team, 2022. The R project for statistical. Computing 3.4.4)[software], R-Project.

Robitaille, A., Saucier, J.-P., 1998. Paysages régionaux du Québec méridional [Regional landscapes of southern Quebec]. Québec: Les publications du Québec. p 213.

Rogers, N.S., D'amato, A.W., Leak, W.B., 2020. Long-term evolution of composition and structure after repeated group selection over eight decades. Can. J. For. Res. 51, 1080–1091. https://doi.org/10.1139/CJFR-2020-0339.

Rosemier, J.N., Storer, A.J., 2010. Assessing the Responses of Native Small Mammals to an Incipient Invasion of Beech Bark Disease through Changes in Seed Production of American Beech (Fagus grandifolia). amid 164, 238–259. https://doi.org/10.1674/0003-0031-164.2.238.

Roy, M.È., Nolet, P., 2018. Early-stage of invasion by beech bark disease does not necessarily trigger American beech root sucker establishment in hardwood stands. Biol. Invasions 20, 3245–3254. https://doi.org/10.1007/s10530-018-1771-7.

Sage, R.W., Porter, W.F., Underwood, H.B., 2003. Windows of opportunity: white-tailed deer and the dynamics of northern hardwood forests of the northeastern US. J. Nat. Conserv. 10, 213–220. https://doi.org/10.1078/1617-1381-00021.

Saucier, J.-P., Grondin, P., Robitaille, A., Gosselin, J., Morneau, C., Richard, P.J.H., Brisson, J., Sirois, L., Leduc, A., Morin, H., Thiffault, E., Gauthier, S., Lavoie, C., Payette, S., 2009. Écologie forestière. In: MultiMondes, É. (Ed.), Manuel De Foresterie. Ordre des ingénieurs forestiers du Québec, Québec, pp. 165–316.

St-Jean, É., Meunier, S., Nolet, P., Messier, C., Achim, A., 2021. Increased levels of harvest may favour sugar maple regeneration over American beech in northern hardwoods. For. Ecol. Manage. 499, 119607 https://doi.org/10.1016/j.foreco.2021.119607.

Takahashi, K., Arii, K., Lechowicz, M.J., 2010. Codominance of Acer saccharum and Fagus grandifolia: The role of Fagus root sprouts along a slope gradient in an old-growth forest. J. Plant Res. 123, 665–674. https://doi.org/10.1007/s10265-010-0312-y.

Therneau, T.M., Lumley, T., Elizabeth, A., Cynthia, C., 2023. survival: Survival Analysis.

Ward, R.T., 1961. Some aspects of regeneration habits of the american beech. Ecology 42, 828–832. https://doi.org/10.2307/1933517.